For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

Etoposide Injection

Fytosid Injection (20mg/ml)

DESCRIPTION:

A clear, almost colourless to pale yellow, solution, free from visible particles. The solution after dilution should appear in clear, almost colourless to pale yellow.

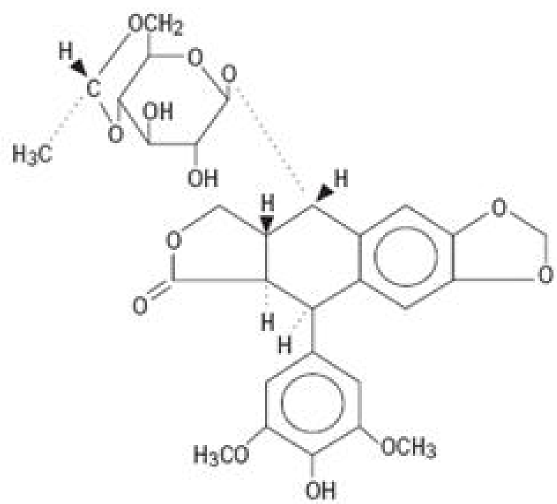
COMPOSITION:

Each ml contains:

Etoposide	20.0 mg
Benzyl Alcohol	30.0 mg
Dehydrated Alcohol	30.5% v/v

CHEMICAL STRUCTURE:

The structural formula of Etoposide is as follows:



The chemical name of etoposide is 4'-demethylepipodophyllotoxin 9-[4,6-O-(R)-ethylidene-β-D-glucopyranoside].

Molecular formula: $C_{29}H_{32}O_{13}$

Relative molecular mass: 588.56.

PHARMACOLOGY:

Mechanism of action

Etoposide is a semisynthetic derivative of podophyllotoxin used in the treatment of certain neoplastic diseases. Podophyllotoxins inhibit mitosis by blocking microtubular assembly. Etoposide inhibits cell cycle progression at a premitotic phase (late S and G2). It does not interfere with the synthesis of nucleonic acids.

Pharmacokinetics

Distribution
The mean volumes of distribution at steady state range from 18 to 29 liters. Etoposide shows low penetration into the CSF. *In vitro*, etoposide is highly protein bound (97%) to human plasma proteins. Etoposide binding ratio correlates directly with serum albumin in cancer patients and normal volunteers. Unbound fraction of etoposide correlates significantly with bilirubin in cancer patients.

Following intravenous infusion, the C_{max} and AUC values exhibit marked intra- and inter-subject variability.

Biotransformation
The hydroxyacid metabolite [4' dimethyl-epipodophyllic acid-9-(4,6 0-ethylidene-β-D-glucopyranoside)], formed by opening of the lactone ring, is found in the urine of adults and children. It is also present in human plasma, presumably as the trans isomer. Glucuronide and/or sulphate conjugates of etoposide are also excreted in human urine. In addition, O-demethylation of the dimethoxy phenol ring occurs through the CYP450 3A4 isoenzyme pathway to produce the corresponding catechol.

Elimination
On intravenous administration, the disposition of etoposide is best described as a biphasic process with a distribution half-life of about 1.5 hours and a terminal elimination half-life ranging from of 4–11 hours. The total body clearance values range from 33 to 48 mL/min or 16–36 ml/minute/m^2 and, like the terminal elimination half-life, are independent of dose over a range of 100-600 mg/m^2. After intravenous administration of 14C etoposide (100 to 124 mg/m^2), mean recovery of radioactivity in the urine was 56% (45% of the dose was excreted as etoposide) and faecal recovery of radioactivity was 44% of the administered dose at 120 hours.

Linearity/non-linearity
Total body clearance and the terminal elimination half-life are independent of dose over a range 100 to 600 mg/m^2. Over the same dose range, the areas under the plasma concentration vs. time curves (AUC) and the maximum plasma concentration (C_{max}) values increase linearly with dose.

Renal impairment
Patients with impaired renal function receiving etoposide have exhibited reduced total body clearance, increased AUC and higher steady state volume of distribution.

Hepatic impairment
In adult cancer patients with liver dysfunction, total body clearance of etoposide is not reduced.

Elderly population
Although minor differences in pharmacokinetic parameters between patients ≤65 years and >65 years of age have been observed, these are not considered clinically significant.

Gender
Although minor differences in pharmacokinetic parameters between genders have been observed, these are not considered clinically significant.

Drug interactions

In a study of the effects of other therapeutic agents on in vitro binding of 14C etoposide to human serum proteins, only phenylbutazone, sodium salicylate, and acetylsalicylic acid displaced protein-bound etoposide at concentrations generally achieved *in vivo*.

INDICATIONS:

• Small cell carcinoma of the lung
• Acute monocytic and myelomonocytic leukaemia
• Hodgkin's disease
• Non-Hodgkin's lymphoma

CONTRAINDICATIONS:

- Hypersensitivity to etoposide, podophyllotoxines or podophyllotoxine-derivatives or to any of the excipients
- Lactation
- Concomitant use of yellow fever vaccine or other live vaccines is contraindicated in immunosuppressed patients.

ADVERSE EFFECTS:

Summary of the safety profile

Dose limiting bone marrow suppression is the most significant toxicity associated with etoposide therapy. In clinicalvstudies in which etoposide was administered as a single agent at a total dose of ≥450 mg/m^2 the most frequent adverse reactions of any severity were leucopenia, neutropenia, anaemia, thrombocytopenia, asthenia, nausea and/or vomiting, alopecia and chills and/or fever.

Tabulated summary of adverse reactions

The following adverse reactions have been reported from etoposide clinical studies and post-marketing experience. These adverse reactions are presented by system organ class and frequency, which is defined by the following categories:

very common (≥1/10), common (≥1/100, <1/10), uncommon (≥1/1,000, <1/100), rare (≥1/10,000, <1/1,000), not known (cannot be estimated from the available data)

System Organ Class	**Frequency**	**Adverse Reaction (MedDRA Terms)**
Infections and infestations	Common	Infection
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Common	Acute leukaemia
Blood and the lymphatic system disorders*	Very common	Myelosuppression*, Leukopenia, thrombocytopenia, neutropenia, anemia
Immune system disorders	Common	Anaphylactic-type reactions**

	Not known	angioedema, bronchospasm
Metabolism and nutrition disorders	Not known	tumour lysis syndrome
Nervous system disorders	Common	Dizziness
	Uncommon	Neuropathy peripheral
	Rare	Seizure*** optic neuritis, cortical blindness transient, neurotoxicities (e.g., somnolence, fatigue)
Cardiac disorders	Common	Myocardial infarction, arrythmia
Vascular disorders	Common	Transient systolic hypotension following rapid intravenous administration, hypertension
Respiratory, thoracic and mediastinal disorders	Rare	Pulmonary fibrosis, interstitial pneumonitis
	Not known	bronchospasm
Gastrointestinal disorders	Very common	Abdominal pain, constipation, nausea and vomiting, anorexia
	Common	Mucositis (including stomatitis and esophagitis), diarrhea
	Rare	Dysphagia, dysgeusia
Hepato-biliary disorders	Very common	Hepatotoxicity, alanine aminotransferase increased, alkaline phosphatase increased, aspartate amino transferase increased, bilirubin increased
Skin and subcutaneous tissue tisorders	Very common	Alopecia, pigmentation
	Common	Rash, urticaria, pruritus
	Rare	Stevens-Johnson syndrome, toxic epidermal necrolysis, radiation recall dermatitis
Reproductive system and breast disorders	Not known	Infertility
General disorders and Administration site conditions	Very common	Asthenia, malaise
	Common	Extravasation****, phlebitis
	Rare	Pyrexia

* Myelosuppression with fatal outcome has been observed. ** Anaphylactic-type reactions can be fatal. ***Seizure is occasionally associated with allergic reactions. **** Complications observed for extravasation included local soft tissue toxicity, swelling, pain, cellulitis, and necrosis including skin necrosis.

Description of selected adverse reactions
In the paragraphs below the incidences of adverse events, given as the mean percent, are derived from studies that utilized single agent etoposide therapy.

Hematological Toxicity:
Myelosuppression with fatal outcome has been reported following administration of etoposide. Myelosuppression is most often dose-limiting. Bone marrow recovery is usually complete by day 20, and no cumulative toxicity has been reported.

Granulocyte and platelet nadirs tend to occur about 10-14 days after administration of etoposide or etoposide phosphate depending on the way of administration and treatment scheme. Nadirs tend to occur earlier with intravenous administration compared to oral administration.

Leukopenia and severe leukopenia (less than 1,000 cells/mm^3) were observed in 91% and 17%, respectively, for etoposide. Thrombocytopenia and severe thrombocytopenia (less than 50,000 platelets/mm^3) were seen in 23% and 9%, respectively, for etoposide. Reports of fever and infection were also very common in patients with neutropenia treated with etoposide. Bleeding has been reported.

Gastrointestinal Toxicity:
Nausea and vomiting are the main gastrointestinal toxicities of etoposide. The nausea and vomiting can usually be controlled by antiemetic therapy.

Alopecia:
Reversible alopecia, sometimes progressing to total baldness has been observed in patients treated with etoposide.

Blood Pressure Changes

Hypotension:
Transient hypotension following rapid intravenous administration has been reported in patients treated with etoposide and has not been associated with cardiac toxicity or electrocardiographic changes. Hypotension usually responds to cessation of infusion of etoposide and/or other supportive therapy as appropriate. When restarting the infusion, a slower administration rate should be used. No delayed hypotension has been noted.

Hypertension**:**
In clinical studies involving etoposide injection, episodes of hypertension have been reported. If clinically significant hypertension occurs in patients receiving etoposide, appropriate supportive therapy should be initiated.

Hypersensitivity:
Anaphylactic-type reactions have also been observed to occur during or immediately after

intravenous administration of etoposide. The role that concentration or rate of infusion plays in the development of anaphylactic-type reactions is uncertain. Blood pressure usually normalizes within a few hours after cessation of the infusion. Anaphylactic reactions can occur with the initial dose of etoposide.

Anaphylactic reactions, manifested by chills, tachycardia, bronchospasm, dyspnoea, diaphoresis, pyrexia, pruritus, hypertension or hypotension, syncope, nausea, and vomiting have been reported to occur in 3% (7 of 245 patients treated with etoposide in 7 clinical studies) of patients treated with etoposide. Facial flushing was reported in 2% of patients and skin rashes in 3%. These reactions have usually responded promptly to the cessation of the infusion and administration of pressor agents, corticosteroids, antihistamines, or volume expanders as appropriate.

Acute fatal reactions associated with bronchospasm have been reported with etoposide. Apnoea with spontaneous resumption of breathing following cessation of infusion have also been reported.

Metabolic Complications:
Tumour lysis syndrome (sometimes fatal) has been observed following the use of episodes in association with other chemotherapeutic drugs.

Paediatric population
The safety profile between paediatric patients and adults is expected to be similar.

DRUG INTERACTIONS:

Effects of other drugs on the pharmacokinetics of etoposide
High dose cyclosporine, resulting in concentrations above 2000 ng/ml, administered with oral etoposide has led to an 80% increase in etoposide exposure (AUC) with a 38% decrease in total body clearance of etoposide compared to etoposide alone.

Concomitant cisplatin therapy is associated with reduced total body clearance of etoposide.

Concomitant phenytoin or phenobarbital therapy is associated with increased etoposide clearance and reduced efficacy, and other enzyme-inducing antiepileptic therapy may be associated with increased etoposide clearance and reduced efficacy.

In vitro, plasma protein binding is 97%. Phenylbutazone, sodium salicylate, and acetylsalicylic acid may displace etoposide from plasma protein binding.

Effect of etoposide on the pharmacokinetics of other drugs
Co-administration of antiepileptic drugs and Etoposide injection can lead to decreased seizure control due to pharmacokinetic interactions between the drugs.

Co-administration of warfarin and etoposide may result in elevated international normalized ratio (INR). Close monitoring of INR is recommended.

Pharmacodynamic interactions
There is increased risk of fatal systemic vaccinal disease with the use of yellow fever vaccine. Live vaccines are contraindicated in immunosuppressed patients.

Prior or concurrent use of other drugs with similar myelosuppressant action as etoposide may be expected to have additive or synergetic effects.

Cross resistance between anthracyclines and etoposide has been reported in preclinical experiments. Paediatric population Interaction studies have only been performed in adults.

PRECAUTIONS AND WARNINGS:
Etoposide should only be administered and monitored under the supervision of a qualified physician experienced in the use of anti-neoplastic medicinal products. In all instances where the use of etoposide is considered for chemotherapy, the physician must evaluate the need and usefulness of the drug against the risk of adverse reactions. Most such adverse reactions are reversible if detected early. If severe reactions occur, the drug should be reduced in dosage or discontinued, and appropriate corrective measures should be taken according to the clinical judgment of the physician.

Reinstitution of etoposide therapy should be carried out with caution, and with adequate consideration of the further need for the drug and close attention to possible recurrence of toxicity.

Myelosuppression
Dose limiting bone marrow suppression is the most significant toxicity associated with etoposide therapy. Fatal myelosuppression has been reported following etoposide administration. Patients being treated with etoposide must be observed for myelosuppression carefully and frequently both during and after therapy. The following haematological parameters should be measured at the start of therapy and prior to each subsequent dose of etoposide: platelet count, haemoglobin, white blood cell count and differential. If radiotherapy or chemotherapy has been given prior to starting etoposide treatment, an adequate interval should be allowed to enable the bone marrow to recover.

Etoposide should not be administered to patients with neutrophil counts less than 1,500 cell/mm^3 or platelet counts less than 100,000 cells/mm^3, unless caused by malignant disease. Doses subsequent to the initial dose should be adjusted if neutrophil count less than 500 cells/mm^3 occurs for more than 5 days or is associated with fever or infection, if platelet count less than 25,000 cells/mm^3 occurs, if any other grade 3 or 4 toxicity develops or if renal clearance is less than 50 ml/min.

Severe myelosuppression with resulting infection or haemorrhage may occur. Bacterial infections should be brought under control before treatment with etoposide.

Secondary leukaemia
The occurrence of acute leukaemia, which can occur with or without myelodysplastic syndrome, has been described in patients that were treated with etoposide containing chemotherapeutic regimens.

Neither the cumulative risk, nor the predisposing factors related to the development of secondary leukaemia are known. The roles of both administration schedules and cumulative doses of etoposide have been suggested but have not been clearly defined.

An 11q23 chromosome abnormality has been observed in some cases of secondary leukaemia in patients who have received epipodophyllotoxins. This abnormality has also been seen in patients developing secondary leukaemia after being treated with chemotherapy regimens not containing epipodophyllotoxins and in leukaemia occurring *de novo*. Another characteristic that has been associated with secondary leukaemia in patients who have received epipodophyllotoxins appears to be a short latency period, with average median time to development of leukaemia being approximately 32 months.

Hypersensitivity
Physicians should be aware of the possible occurrence of an anaphylactic reaction with etoposide, manifested by chills, fever, tachycardia, bronchospasm, dyspnoea and hypotension, which can be fatal. Treatment is symptomatic. The infusion should be terminated immediately, followed by the administration of pressor agents, corticosteroids, antihistamines, or volume expanders at the discretion of the physician.

Hypotension
Etoposide should be given only by slow intravenous infusion (usually over a 30 to 60 minute period) since hypotension has been observed as a possible side effect of rapid intravenous injection.

Injection site reaction
Injection site reactions may occur during administration of etoposide. Given the possibility of extravasation, it is recommended to closely monitor the infusion site for possible infiltration during drug administration.

Low serum albumin
Low serum albumin is associated with increased exposure to etoposide. Therefore patients with low serum albumin may be at increased risk for etoposide-associated toxicities.

Impaired renal function
In patients with moderate (CrCl =15 to 50 mL/min), or severe (CrCl <15ml/min) renal impairment undergoing haemodialysis, etoposide should be administered at a reduced dose.

Haematological parameters should be measured and dose adjustments in subsequent cycles considered based on haematological toxicity and clinical effect in moderate and severe renal impaired patients.

Impaired hepatic function
Patients with impaired hepatic function should regularly have their hepatic function monitored due to the risk of accumulation of etoposide.

Tumour lysis syndrome
Tumour lysis syndrome (sometimes fatal) has been reported following the use of etoposide in association with other chemotherapeutic drugs. Close monitoring of patients is needed to detect early signs of tumour lysis syndrome, especially in patients with risk factors such as bulky treatment-sensitive tumours, and renal insufficiency. Appropriate preventive measures should also be considered in patients at risk of this complication of therapy.

Mutagenic potential
Given the mutagenic potential of etoposide, an effective contraception is required for both male and female patients during treatment and up to 6 months after ending treatment. Genetic consultation is recommended if the patient wishes to have children after ending the treatment. As etoposide may decrease male fertility, preservation of sperm may be considered for the purpose of later fatherhood.

Excipient (s) that the clinician should be aware of:

Ethanol

Etoposide Injection contains alcohol (ethanol). There is a health risk to hepatic patients, alcoholics, epileptics, patients with organic brain diseases, pregnant women, breastfeeding women, and children, amongst others. The effect of other drugs may be reduced or increased.

Benzyl alcohol

Etoposide Injection contains benzyl alcohol. Benzyl alcohol may cause allergic reactions. Benzyl alcohol has been linked with the risk of severe side effects including breathing problems (called "gasping syndrome") in young children.
Should not be given to newborn babies (up to 4 weeks old).
Should not be used for more than a week in young children (less than 3 years old).
Caution should be exercised in pregnant or breast-feeding patients or if the patient has a liver or kidney disease. This is because large amounts of benzyl alcohol can build-up in the body and may cause side effects (called "metabolic acidosis").

Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. Etoposide may cause adverse reactions that affect the ability to drive or use machines such as fatigue, somnolence, nausea, vomiting, cortical blindness, hypersensitivity reactions with hypotension. Patients who experience such adverse reactions should be advised to avoid driving or using machines.

PREGNANCY AND LACTATION:

Pregnancy

There are no or limited amount of data from the use of etoposide in pregnant women. Studies in animals have shown reproductive toxicity. In general etoposide can cause fetal harm when administered to pregnant women. Etoposide injection should not be used during pregnancy unless the clinical condition of the woman requires treatment with etoposide. Women of childbearing potential should be advised to avoid becoming pregnant. Women of childbearing potential have to use effective contraception during and up to 6 months after treatment. If the drug is used during pregnancy, or if the patient becomes pregnant while receiving the drug, the patient should be informed of the potential hazard to the fetus.

Breastfeeding

Etoposide is excreted in human milk. There is the potential for serious adverse reactions in nursing infants from etoposide. A decision must be made whether to discontinue breast-feeding or to discontinue Etoposide injection, taking into account the benefitof breast feeding for the child and the benefit of therapy for the woman.

Benzyl alcohol is probably excreted into breast milk and can be orally absorbed by the infant.

Fertility

As etoposide may decrease male fertility, preservation of sperm may be considered for the purpose of later fatherhood.

Women of childbearing potential/Contraception in males and females
Women of childbearing potential should use appropriate contraceptive measures to avoid pregnancy during etoposide therapy. Etoposide has been shown to be teratogenic in mice and rats. Given the mutagenic potential of etoposide, an effective contraceptive is required for both male and female patients during treatment and up to 6 months after ending treatment. Genetic consultation is recommended if the patient wishes to have children after ending treatment.

DOSAGE AND ADMINISTRATION:

Dosage
The usual dose of etoposide must be based on the clinical and haematological response and tolerance of the patient. A repeat course of etoposide should not be administered until the patient's haematological function is within acceptable limits.

Adult
The dosage for Etoposide Injection is 50-60 mg/m^2/day intravenously for 5 consecutive days followed by a treatment free interval of 2-3 weeks. Total dose should not usually exceed 400 mg/m^2 per course.

Method of administration
Plastic devices made of acrylic or ABS (a polymer of acrylonitrile, butadiene and styrene) have been reported to crack or leak when used with undiluted Etoposide injection.

Etoposide should only be given by slow intravenous infusion.

Etoposide should not be administered by intrapleural or intraperitoneal injection.

Etoposide must be diluted before administration. Resultant concentrations should not be greater than 0.4 mg/mL since precipitation can occur. Usually etoposide is added to 250 mL of 0.9% sodium chloride or 5% glucose. The infusion should be administered over a period of 30-60 minutes.

Contact with buffered aqueous solutions with pH above 8 should be avoided. Concentrations of etoposide of 0.4 mg/mL in 5% glucose or 0.9% sodium chloride are chemically stable for 24 hours when stored at room temperature. However, to reduce microbiological hazard, admixtured solutions should be used as soon as practicable after preparation. If storage is required hold at 2-8°C for no more than 24 hours. Contains no antimicrobial preservative, use once only and discard any residue.

The use of a Pharmacy Bulk Pack should be restricted to suitably qualified pharmacists operating in suitably equipped hospital pharmacies or compounding centres. The Pharmacy Bulk Pack is intended for multiple dispensing into sterile solutions for subsequent infusion in individual patients in one treatment session. The Pharmacy Bulk Pack should be spiked only once.

Dosage Adjustment

Hepatic impairment:
Etoposide is contraindicated in severe hepatic dysfunction, and it should be used with caution in patients with mild to moderate hepatic impairment.

Renal impairment:
Since some etoposide (approximately 30%) is excreted unchanged in the urine, dosage adjustment may be needed in patients with impaired renal function.

OVERDOSE:
Total etoposide doses of 2.4 g/m² to 3.5 g/m² administered intravenously over three days have resulted in severe mucositis and myelotoxicity. Metabolic acidosis and cases of severe hepatic toxicity have been reported in patients receiving higher than recommended intravenous doses of etoposide.

There is no specific antidote available. Treatment should therefore be symptomatic and supportive, and patients should be closely monitored. Etoposide and its metabolites are not dialyzable.

STORAGE:
Store below 30°C. Protect from light. Do not freeze.
For single use only. Discard any remaining solution.

After dilution:
Chemical and physical in-use stability of the solution diluted to a concentration of 0.2 mg/ml or 0.4mg/ml with 0.9% sodium chloride or 5% dextrose solution has been demonstrated up to 96 hours at 30°C.

From a microbiological point of view, the diluted product should be used immediately. When the diluted product is stored in PVC bags a leaching of plasticizer in the product or precipitation may occur. Therefore, it is recommended to use glass containers for the diluted product.

INCOMPATIBILITIES:
Plastic devices made of acrylic or ABS polymers have been observed to crack when used with undiluted etoposide. This effect has not been observed with etoposide after dilution of the concentrate for solution for infusion according to instructions.

This medicinal product must not be mixed with other medicinal product except those mentioned in the Special Precautions for Handling section.

PACKAGING:
Fytosid Injection (20mg/ml) is available as 5 ml injection containing 100 mg of etoposide.

SPECIAL PRECAUTIONS FOR HANDLING:
Etoposide is administered by slow intravenous infusion.
Etoposide SHOULD NOT BE GIVEN BY RAPID INTRAVENOUS INJECTION.

Etoposide must be diluted immediately prior to use with either 50 mg/ml (5%) dextrose in water, or 9 mg/ml (0.9%) sodium chloride solution to give a final concentration of 0.2 mg/ml to 0.4 mg/ml.

At higher concentrations precipitation of etoposide may occur. Solutions showing any signs of precipitation should not be used.

Administration precautions

Hypotension following rapid intravenous administration has been observed. Hence it is recommended that the etoposide solution be administered over a 30- to 60- minute period. Longer infusion times may be required based on patient tolerance. As with other potentially toxic compounds, caution should be exercised in handling and preparing the solution of etoposide. Skin reactions associated with accidental exposure to etoposide may occur. The use of gloves is recommended. If etoposide solution contacts the skin or mucosa, immediately wash the skin or mucosa thoroughly with soap and water.

Cytotoxic should not be handled by pregnant personnel. Waste-disposal procedures should take into account the cytotoxic nature of this substance.

MANUFACTURER INFORMATION:

Manufactured in India by:
Fresenius Kabi Oncology Ltd.
Village - Kishanpura,
P.O. Guru Majra,
Tehsil - Nalagarh,
Distt. Solan, (H.P.) - 174 101

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